

Task 1 : Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Term	Full Form	One-line Use
HTTP	HyperText Transfer Protocol	Used to transfer web pages and data between a client and server.
HTTPS	HyperText Transfer Protocol Secure	Securely transfers web data using encryption.
FTP	File Transfer Protocol	Used to transfer files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used to send emails from one mail server to another.

Task 2 : What is the difference between IPv4 and IPv6? Answer in 2 lines.

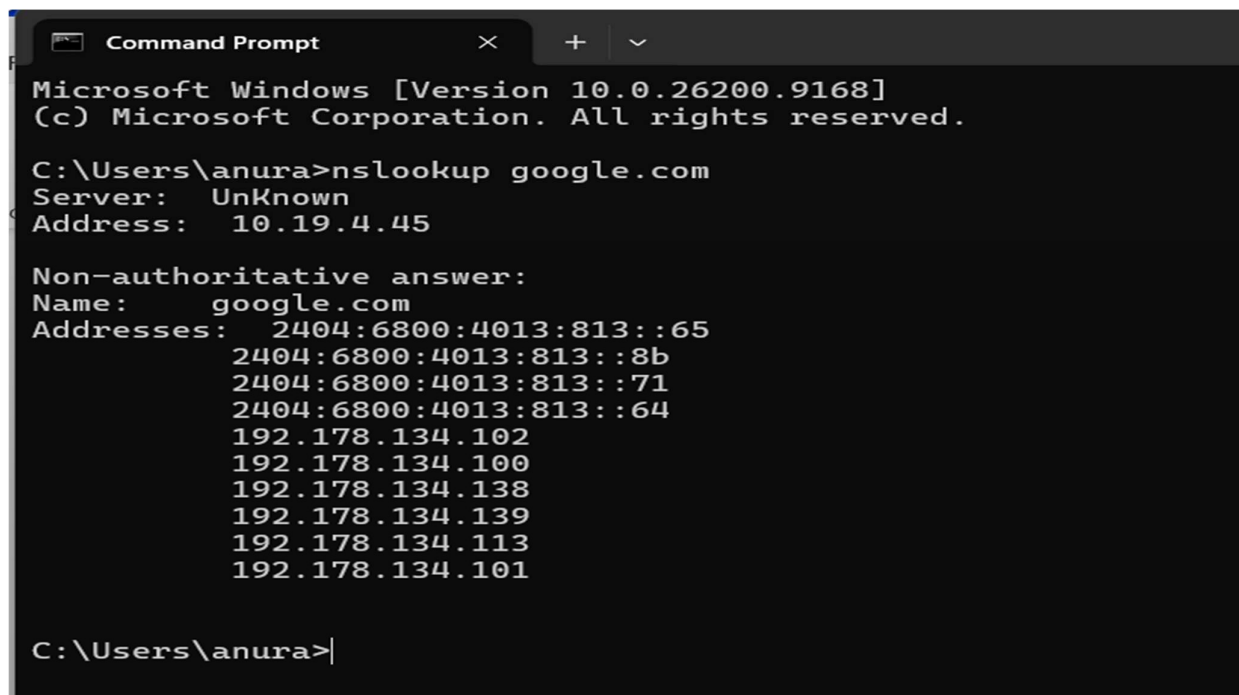
IPv4

uses a 32-bit address and provides about 4.3 billion addresses, e.g., 192.168.1.1.

IPv6

uses a 128-bit address and provides a much larger number of addresses, e.g., 2001:db8::1.

Task 3 : Run nslookup google.com on your system and write down the IP address you get.



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Command Prompt
Microsoft Windows [Version 10.0.26200.9168]
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C:\Users\anura>nslookup google.com
Server:    UnKnown
Address:   10.19.4.45

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4013:813::65
          2404:6800:4013:813::8b
          2404:6800:4013:813::71
          2404:6800:4013:813::64
          192.178.134.102
          192.178.134.100
          192.178.134.138
          192.178.134.139
          192.178.134.113
          192.178.134.101

C:\Users\anura>
```

Task 4 : What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Code	Meaning	Explanation
200	OK	Request was successful.
301	Moved Permanently	The requested resource has permanently moved to another URL.
404	Not Found	The requested page or resource could not be found.
500	Internal Server Error	An unexpected error occurred on the server.
403	Forbidden	The server understood the request but refuses to allow access.

Task 5 : Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

DNS hierarchy works from **Root → TLD → Domain → Subdomain**.

For www.google.com:

- **Root (.)** → The top level of the DNS hierarchy.
- **TLD (.com)** → The Top-Level Domain.
- **Domain (google)** → The registered domain name.
- **Subdomain (www)** → A subdomain/hostname used to identify a particular service or web server.

So, the hierarchy is:

Root (.) → TLD (com) → Domain (google) → Subdomain (www).
